Learning Theory Bounds for Strategic Voluntary Disclosure (0-1 Loss + Rademacher Complexity)

Abstract

We derive non-asymptotic generalization bounds for the static learning problem in multi-class strategic voluntary disclosure under 0-1 loss. The bounds are obtained via Rademacher complexity and VC-dimension arguments, properly accounting for random subsample sizes and the combinatorial choice over K possible defaults. All steps are shown in full detail.

1 Setup and Notation

Let $Y, U \in \{1, \ldots, K\}$ and $X \in \mathbb{R}^d$. We observe n i.i.d. samples $S = \{(u_j, x_j, y_j)\}_{j=1}^n$.

For each default $i = 1, \ldots, K$ we learn a linear multiclass predictor

$$h_i(x) = \arg\max_{c=1}^{K} \theta_{i,c}^\top x, \quad \|\theta_i\|_\infty \le C.$$

The hypothesis class for default i is

$$\mathcal{H}_i = \{x \mapsto \arg\max_c \theta_c^\top x \mid \|\theta\|_\infty \le C\}.$$

The corresponding 0-1 loss class is

$$\mathcal{L}_i = \{(x, y) \mapsto \mathbb{I}\{h(x) \ne y\} \mid h \in \mathcal{H}_i\}.$$

The **true strategic risk** for default i and predictor h is

$$R_S^i(h, q) = P(Y \ne i) + (1 - \pi_i)(1 - q)\big(r_X^{(-i)}(h) - P(Y \ne i \mid U \ne i)\big),$$

where $r_X^{(-i)}(h)$ is the 0-1 risk of h on the subpopulation $U \ne i$.

The **empirical strategic risk** is the plug-in estimator

$$\hat{R}_S^i(h, q) = \frac{1}{n} \sum_{j=1}^{n} \ell_{0-1}(y_j, \hat{y}(M_j; i, h), q).$$

2 Learning Procedure

For each default $i = 1, \ldots, K$ we solve the empirical risk minimization problem

$$\hat{h}_i = \arg\min_{h \in \mathcal{H}_i} \hat{R}_S^i(h, q).$$

(Note that samples with $u_j = i$ contribute a constant and can be ignored; only samples with $u_j \ne i$ are used for training $\hat{h}_i$.)

After obtaining the K predictors $\hat{h}_1, \ldots, \hat{h}_K$, we select the empirically best default

$$\hat{i}^* = \arg\min_{i=1,\ldots,K} \hat{R}_S^i(\hat{h}_i, q).$$

The output strategy is the pair $(\hat{i}^*, \hat{h}_{\hat{i}^*})$.

3 Handling Random Subsample Sizes

For each i let $n_i = |\{j : u_j \neq i\}|$ be the (random) number of revealed samples for default i.

Lemma 1 (Safe Sample Size)**.** *With probability at least* $1 - \delta/K$*, simultaneously for all* $i = 1, \dots, K$*,*

$$n_i \geq \frac{n}{2}.$$

(Here we assume $\min_i(1 - \pi_i) \geq 1/2$*; the constant can be adjusted for general* π_i*.)*

Proof. Each n_i is Binomial$(n, 1 - \pi_i)$. By Hoeffding's inequality and a union bound over the K defaults, the probability that any n_i falls below half its expectation is at most δ/K. □

On this high-probability event we may safely replace every occurrence of n_i by $n/2$ in the complexity terms below.

4 Uniform Convergence for a Fixed Default

Lemma 2 (Rademacher Uniform Convergence for Fixed Default i)**.** *Let* $\hat{h}_i$ *be the empirical risk minimizer for default* i*. Then, with probability at least* $1 - \delta/K$*,*

$$\left| R_S^i(\hat{h}_i, q) - \hat{R}_S^i(\hat{h}_i, q) \right| \leq 2\,\mathrm{Rad}_{n_i}(\mathcal{L}_i) + O\left(C \sqrt{\frac{\log(K/\delta)}{n}} \right).$$

Proof. **Step 1 (Symmetrization).** Introduce an independent ghost sample S' and i.i.d. Rademacher variables $\sigma_j = \pm 1$. By the standard symmetrization argument,

$$\mathbb{E}\Big[\sup_{h \in \mathcal{H}_i} \left| R_S^i(h, q) - \hat{R}_S^i(h, q) \right| \Big] \leq 2\,\mathrm{Rad}_{n_i}(\mathcal{L}_i),$$

where $\mathrm{Rad}_{n_i}(\mathcal{L}_i)$ is the empirical Rademacher complexity of the 0-1 loss class $\mathcal{L}_i$.

Step 2 (VC Dimension of the Loss Class). The class $\mathcal{L}_i$ consists of 0-1 loss functions of linear multiclass predictors with K classes. Its VC dimension satisfies

$$\mathrm{VC}(\mathcal{L}_i) \leq O(Kd)$$

(by the Sauer-Shelah lemma applied to linear threshold functions in dimension d with K labels).

Step 3 (Rademacher Complexity Bound). By Massart's lemma combined with the Sauer-Shelah bound,

$$\mathrm{Rad}_{n_i}(\mathcal{L}_i) \leq O\left(\sqrt{\frac{Kd \log n_i}{n_i}} \right).$$

On the safe-sample-size event of Lemma 1 we have $n_i \geq n/2$, therefore

$$\mathrm{Rad}_{n_i}(\mathcal{L}_i) \leq O\left(\sqrt{\frac{Kd \log n}{n}} \right).$$

Step 4 (Concentration). The loss is bounded in $[0, 1]$. By McDiarmid's bounded-difference inequality the deviation around the expectation concentrates with probability $1 - \delta/K$ at rate $O(\sqrt{\log(K/\delta)/n})$. Combining with the expectation bound yields the stated result. □

5 Simultaneous Bound for All Defaults

Lemma 3 (Simultaneous Uniform Convergence)**.** *With probability at least $1-\delta$, the following holds simultaneously for all defaults $i = 1, \ldots, K$:*

$$\left|R_S^i(\hat{h}_i, q) - \hat{R}_S^i(\hat{h}_i, q)\right| \le O\left(\sqrt{\frac{Kd\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

Proof. Apply Lemma 2 to each of the K defaults and take a union bound. The failure probability per default is δ/K, so the total failure probability is at most δ. □

6 Generalization Bound for Strategic Advantage

Theorem 1 (Strategic Advantage Generalization)**.** *Let $\Delta^*(q) = R_V^*(q) - R_S^*(q)$ be the true strategic advantage and let $\hat{\Delta}(q)$ be the empirical advantage of the learned strategy $(\hat{i}^*, \hat{h}_{\hat{i}^*})$. Then, with probability at least $1-\delta$,*

$$\left|\Delta^*(q) - \hat{\Delta}(q)\right| \le O\left(\sqrt{\frac{K^2 d\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

Proof. The vanilla risk $R_V^*(q)$ admits a standard uniform convergence bound of order $O(\sqrt{(d\log n)/n})$ (ordinary multiclass classification). For the strategic risk we have, from Lemma 3,

$$R_S^*(q) = \min_i R_S^i(\hat{h}_i, q), \qquad \hat{R}_S^{\hat{i}^*}(q) = \min_i \hat{R}_S^i(\hat{h}_i, q).$$

The elementary inequality $|\min_i a_i - \min_i b_i| \le \max_i |a_i - b_i|$ together with the simultaneous bound of Lemma 3 yields

$$\left|R_S^*(q) - \hat{R}_S^{\hat{i}^*}(q)\right| \le O\left(\sqrt{\frac{K^2 d\log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

Subtracting the vanilla bound (which has the same order, without the extra K) gives the claimed result for the advantage $\Delta^*(q)$. □

7 Excess Risk of the Learned Strategy

Theorem 2 (Excess Strategic Risk)**.** *With probability at least $1-\delta$,*

$$R_S^{\hat{i}^*}(q) - R_S^*(q) \le O\left(\sqrt{\frac{K^2 d\log n}{n}}\right).$$

Proof. The excess risk is bounded by the deviation of the selected empirical strategy from the best-in-hindsight strategy. By the same argument as in Theorem 1 (using the simultaneous bound of Lemma 3 and the min-operator inequality), the deviation is at most the maximum per-default deviation, which is of the stated order. □

8 Summary of Main Results

- Per-default uniform convergence (Lemma 2): $O(\sqrt{(Kd\log n)/n})$.
- Simultaneous bound over all K defaults (Lemma 3): same order.

- Strategic advantage generalization (Theorem 1): $O(\sqrt{(K^2 d \log n)/n})$.
- Excess risk of output strategy (Theorem 2): $O(\sqrt{(K^2 d \log n)/n})$.

All constants hidden in the $O(\cdot)$ notation are universal (independent of the underlying distribution) and can be made fully explicit if desired.